Unit ID: 6739f3daa34577ff4ce159f4

Unit 2 First Steps with MODI: Unplugged Activities

PDF Documents

PDF Document 1: 1731851225755-Chapter 2_unit2 First Steps with MODI Unplugged Activities.pdf

This PDF document is included in the unit materials.

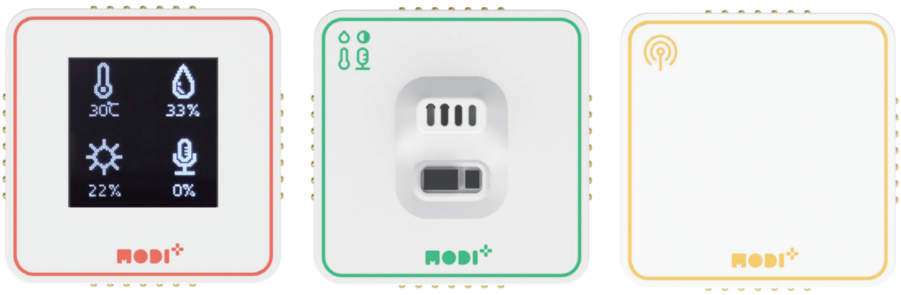
UNIT 2 FIRST STEPS WITH MODI: UNPLUGGED ACTIVITIES

In the world of robotics and programming, an "unplugged activity" is a bit like a hands-on puzzle or game. It doesn't need computers or electronic devices to work.

Before we start playing with our MODI modules, we're going to warm up our brains with some unplugged activities. This will help us think like programmers and understand the steps our MODI robots will need to follow when we start creating with them.

UNPLUGGED

Activity 1 Environment Meter



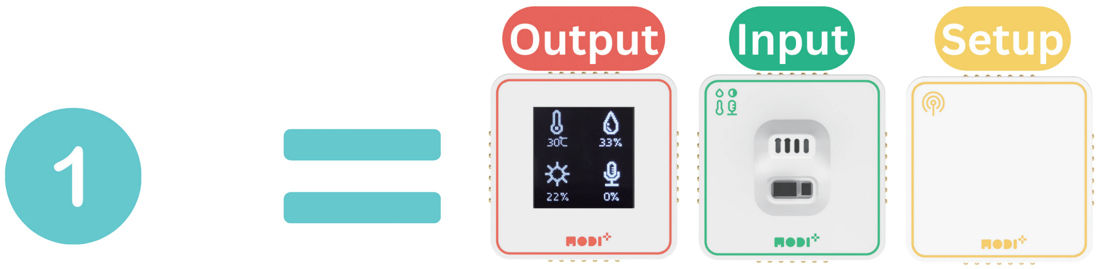
Instructions:

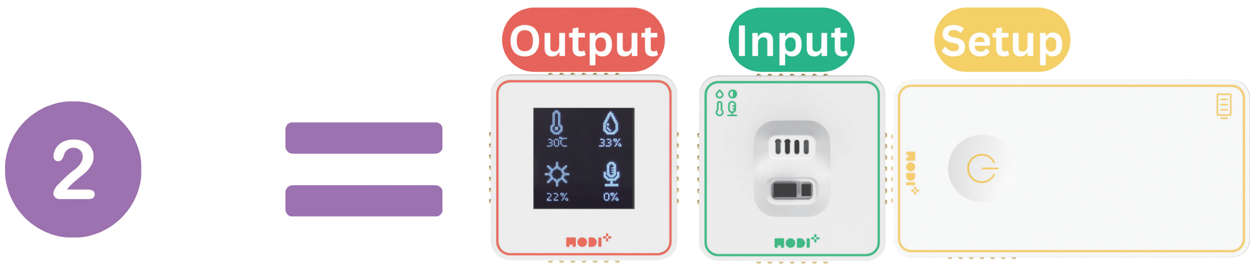
1) **Connecting Your MODI Modules:** Just bring your modules close together, and they'll snap into place like magic, thanks to the magnets on their sides.

2) **Powering Up Your Modules:** There are two cool ways to make your modules come alive:

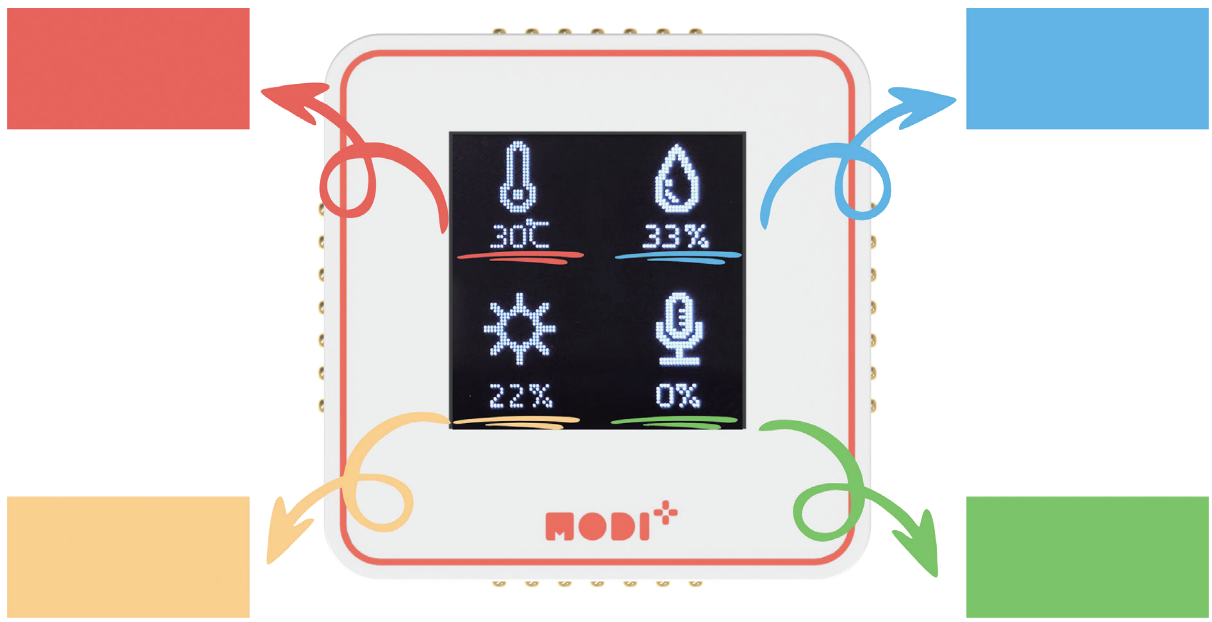
a. Use the Network module to connect to your PC.

b. Use the Battery module for power for the connected modules.





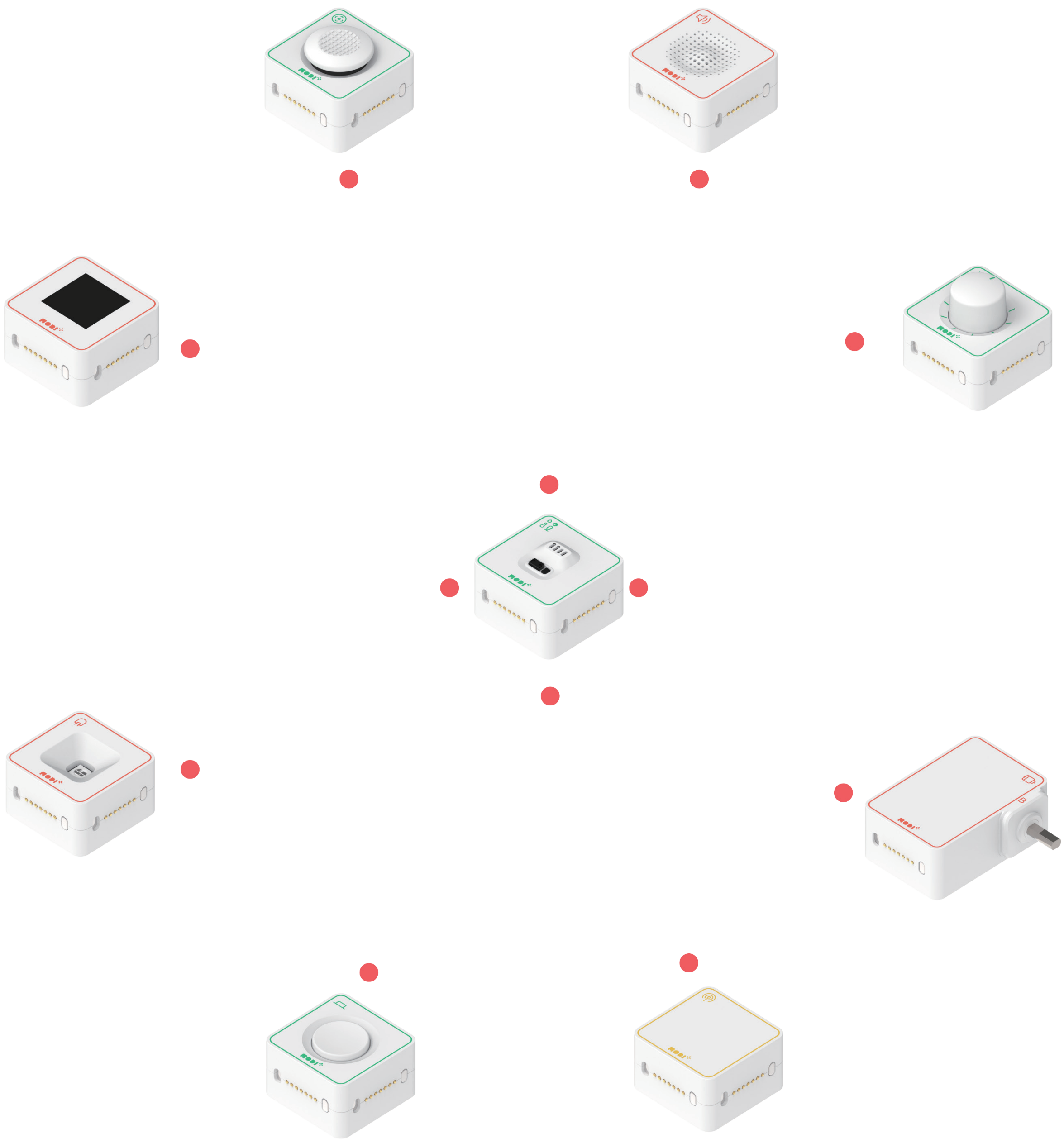
3) **Checking the Display:** Your Display module will show 4 different types of information. The information is being transmitted from the Environment module to the Display module. **Identify** each of the information *(among temperature, humidity, illuminance, and sound level)*, and fill in the blanks below.

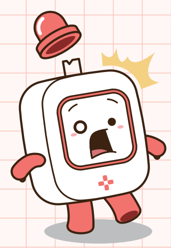


Activity 2 Connecting Worlds: Environment and Output

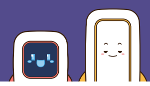
Instructions:

Find 4 MODI modules that can interact with the Environment Module and connect them.





Remember, MODI modules function similarly to components in robotics and computing systems. Ensure your connections between the Environment Module and other modules follow **an "input" to "output" relationship.**



In the blanks below, list the modules you connected to the Environment Module in the "Connecting Worlds" activity. Describe how each pair functions and find similar devices we use every day that work the same way.

Environment + Motor A

How it works:

The Environment Module senses changes in its surroundings, such as light or temperature, and based on these changes, it signals Motor A to start moving.

Examples in daily life:

Automatic curtains or blinds operate in our homes, where they open or close in response to sunlight levels detected by a sensor.

Environment +

Environment +

Environment +